

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250284355

Kind Code

A1

Publication Date

September 11, 2025

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Display Panel and Display Device Comprising the Same

Abstract

Disclosed is a display panel and a display device capable of reducing the reflection of external light. The display panel includes a touch sensor, disposed on an encapsulation unit, and an antireflective film disposed in a non-emission area of the touch sensor so as to overlap each other, whereby it is possible to reduce external light from being incident on the touch sensor and a routing line, and therefore it is possible to reduce the reflectance of external light without using an expensive polarizing plate.

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Family ID: 1000008619713

Appl. No.: 19/215003

Filed: May 21, 2025

Foreign Application Priority Data

KR 10-2018-0168152

Dec. 24, 2018

Related U.S. Application Data

parent US continuation 18543671 20231218 parent-grant-document US 12336383 child US 19215003

parent US continuation 18166843 20230209 parent-grant-document US 11943967 child US 18543671

parent US continuation 17408001 20210820 parent-grant-document US 11611060 child US 18166843

parent US continuation 16707470 20191209 parent-grant-document US 11127929 child US 17408001

Publication Classification

Int. Cl.: **G06F3/041** (20060101); **H10K50/86** (20230101); **H10K59/122** (20230101); **H10K59/40** (20230101); **H10K59/80** (20230101)

U.S. Cl.:

CPC **G06F3/0412** (20130101); **G06F3/041** (20130101); **H10K50/865** (20230201); **H10K59/122** (20230201); **H10K59/40** (20230201); **H10K59/805** (20230201); **H10K59/873** (20230201); **H10K59/8792** (20230201); G06F2203/04112 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/543,671 filed on Dec. 18, 2023, which is a continuation of U.S. patent application Ser. No. 18/166,843 filed on Feb. 9, 2023, which is a continuation of U.S. patent application Ser. No. 17/408,001 filed on Aug. 20, 2021, which is a continuation of U.S. patent application Ser. No. 16/707,470 filed on Dec. 9, 2019, which claims the benefit of Republic of Korea Patent Application No. 10-2018-0168152, filed on Dec. 24, 2018, each of which is hereby incorporated by reference in its entirety.

BACKGROUND

Field of Technology

[0002] The present disclosure relates to a display panel and a display device, and more particularly to a display panel and a display device capable of reducing the reflection of external light.

Discussion of the Related Art

[0003] Image display devices, which are a core technology in the information and communication age and serve to display various kinds of information on a screen, have been developed such that the image display devices are thinner, lighter, and portable and exhibit high performance. As a result, flat panel display devices, which have lower weight and volume than cathode ray tubes (CRT), have received a great deal of attention.

[0004] Representative examples of such flat panel display devices may include a liquid crystal display (LCD) device, a plasma display panel (PDP) device, an organic light-emitting display (OLED) device, and an electrophoretic display (ED) device.

[0005] A flat panel display device includes a plurality of signal lines and a plurality of driving electrodes. External light is reflected by the signal lines and the driving electrodes, whereby external visibility is reduced.

[0006] In the case in which a polarizing plate is used in order to solve this problem, it is possible to reduce the reflection of external light. However, transmittance is reduced and price competitiveness is deteriorated due to the polarizing plate, which is expensive.

SUMMARY

[0007] Accordingly, the present disclosure is directed to a display panel and a display device that substantially obviates one or more problems due to limitations and disadvantages of the related art.

[0008] An object of the present disclosure is to provide a display panel and a display device capable of preventing the reflection of external light.

[0009] Additional advantages, objects, and features of the invention will be set forth in part in the description which follows and in part will become apparent to those having ordinary skill in the art upon examination of the following or may be learned from practice of the invention. The objectives and other advantages of the invention may be realized and attained by the structure particularly

pointed out in the written description and claims hereof as well as the appended drawings.

[0010] To achieve these objects and other advantages and in accordance with the purpose of the invention, as embodied and broadly described herein, a display panel includes a touch sensor, disposed on an encapsulation unit, and an antireflective film disposed in a non-emission area of the touch sensor so as to overlap each other, whereby it is possible to prevent external light from being incident on the touch sensor and a routing line, and therefore it is possible to reduce the reflectance of external light without using an expensive polarizing plate.

[0011] It is to be understood that both the foregoing general description and the following detailed description of the present invention are exemplary and explanatory and are intended to provide further explanation of the invention as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this application, illustrate embodiment(s) of the invention and together with the description serve to explain the principle of the invention. In the drawings:

[0013] FIG. 1 is a perspective view showing an organic light-emitting display panel having a touch sensor according to one embodiment of the present disclosure;

[0014] FIG. 2 is a plan view of the organic light-emitting display panel having the touch sensor shown in FIG. 1 according to one embodiment of the present disclosure;

[0015] FIG. 3 is a sectional view of the organic light-emitting display panel taken along line I-I' of FIG. 2 according to one embodiment of the present disclosure;

[0016] FIG. 4 is a detailed plan view showing the touch electrode shown in FIG. 2 according to one embodiment of the present disclosure;

[0017] FIG. 5 is a sectional view of the organic light-emitting display panel taken along line II-II' of FIG. 4 according to one embodiment of the present disclosure;

[0018] FIG. 6 is a sectional view showing another embodiment of the black layer shown in FIG. 5 according to one embodiment of the present disclosure;

[0019] FIG. 7 is a view showing reflectance based on the content of white particles included in the white layer shown in FIG. 5 according to one embodiment of the present disclosure;

[0020] FIG. 8 is a view illustrating a process of reducing the reflection of external light in an organic light-emitting display panel having a black layer and a white layer according to one embodiment of the present disclosure; and

[0021] FIGS. 9A and 9B are sectional views showing an organic light-emitting display panel having a cover substrate according to one embodiment of the present disclosure.

DETAILED DESCRIPTION

[0022] Reference will now be made in detail to embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

[0023] FIG. 1 is a perspective view showing an organic light-emitting display panel having a touch sensor according to the present disclosure, and FIG. 2 is a plan view of the organic light-emitting display panel having the touch sensor according to the present disclosure.

[0024] The organic light-emitting display panel having the touch sensor shown in FIGS. 1 and 2 includes a plurality of subpixels PXL arranged on a substrate 111 in a matrix fashion, an encapsulation unit 140 disposed on the subpixels PXL, and mutual capacitance Cm disposed on the encapsulation unit 140.

[0025] The organic light-emitting display panel having the touch sensor displays an image through

the subpixels PXL, each of which includes a light-emitting element **120**, during a display period. In addition, the organic light-emitting display panel having the touch sensor detects variation in mutual capacitance C_m (the touch sensor) due to a user's touch during a touch period to sense whether a touch has been performed and the touched position.

[0026] Each of the subpixels PXL, which are disposed in an active area of the organic light-emitting display panel having the touch sensor, includes a pixel-driving circuit and a light-emitting element **120** connected to the pixel-driving circuit.

[0027] As shown in FIG. **1**, the pixel-driving circuit includes a switching transistor **T1**, a driving transistor **T2**, and a storage capacitor C_{st} . Meanwhile, in the present disclosure, the pixel-driving circuit has been described as including two transistors **T** and one capacitor **C** (2T1C) by way of example. However, the present disclosure is not limited thereto. That is, a 3T1C or 3T2C type pixel-driving circuit having three or more transistors **T** and one or more capacitors **C** may be used.

[0028] When a scan pulse is supplied to a scan line **SL**, the switching transistor **T1** is turned on to supply a data signal, which is supplied to a data line **DL**, to the storage capacitor C_{st} and to a gate electrode of the driving transistor **T2**.

[0029] In response to the data signal supplied to the gate electrode of the driving transistor **T2**, the driving transistor **T2** controls the current that is supplied from a high-voltage (**VDD**) supply line to the light-emitting element **120** to adjust the amount of light emitted by the light-emitting element **120**. Even when the switching transistor **T1** is turned off, the driving transistor **T2** supplies uniform current to the light-emitting element **120** using the voltage charged in the storage capacitor C_{st} such that the light-emitting element **120** keeps emitting light until a data signal of the next frame is supplied.

[0030] To this end, as shown in FIG. **3**, the driving transistor **T2** includes a semiconductor layer **134** disposed on a buffer layer **112**, a gate electrode **132** overlapping the semiconductor layer **134** in the state in which a gate dielectric film **102** is disposed therebetween, and source electrode **136** and drain electrode **138** formed on an interlayer dielectric film **114** so as to contact the semiconductor layer **134**.

[0031] The semiconductor layer **134** is made of at least one of an amorphous semiconductor material, a polycrystalline semiconductor material, or an oxide semiconductor material. The semiconductor layer **134** includes a channel area, a source area, and a drain area. The channel area overlaps the gate electrode **132** in the state in which the gate dielectric film **102** is disposed therebetween so as to be formed between the source and drain electrodes **136** and **138**. The source area is electrically connected to the source electrode **136** via a source contact hole, which is formed through the interlayer dielectric film **114**. The drain area is electrically connected to the drain electrode **138** via a drain contact hole, which is formed through the interlayer dielectric film **114**.

[0032] The gate electrode **132** may be made of one of molybdenum (**Mo**), aluminum (**Al**), chrome (**Cr**), gold (**Au**), titanium (**Ti**), nickel (**Ni**), neodymium (**Nd**), or copper (**Cu**) or an alloy thereof, and may have a single-layered structure or a multi-layered structure. However, the present disclosure is not limited thereto.

[0033] The gate electrode **132** overlaps the channel area of the semiconductor layer **134** in the state in which the gate dielectric film **102** is disposed therebetween. At this time, as shown in FIG. **3**, the gate dielectric film **102** may be formed so as to have the same line width as the gate electrode **132** in order to expose the side surface of the semiconductor layer **134**, or may be formed so as to have a larger line width than the gate electrode **132** in order to cover the side surface of the semiconductor layer **134**.

[0034] Each of the source and drain electrodes **136** and **138** may be made of one of molybdenum (**Mo**), aluminum (**Al**), chrome (**Cr**), gold (**Au**), titanium (**Ti**), nickel (**Ni**), neodymium (**Nd**), or copper (**Cu**) or an alloy thereof, and may have a single-layered structure or a multi-layered structure. However, the present disclosure is not limited thereto. The source electrode **136** is connected to the source area of the semiconductor layer **134**, which is exposed through the source

contact hole formed through both the gate dielectric film **102** and the interlayer dielectric film **114** or only through the interlayer dielectric film **114**. The drain electrode **138** faces the source electrode, and is connected to the drain area of the semiconductor layer **134** via the drain contact hole, which is formed through both the gate dielectric film **102** and the interlayer dielectric film **114** or only through the interlayer dielectric film **114**.

[0035] The light-emitting element **120** includes an anode electrode **122**, at least one light-emitting stack **124** formed on the anode electrode **122**, and a cathode electrode **126** formed on the light-emitting stack **124**.

[0036] The anode electrode **122** is electrically connected to the drain electrode **138** of the driving transistor T2, which is exposed through a pixel contact hole **116** formed through a passivation film **108** and a planarization layer **118** disposed on the driving transistor T2. The anode electrode **122** of each subpixel is formed so as to be exposed by a bank **128**. The bank **128** is formed so as to expose the anode **132**. The bank **128** may be made of an opaque material (e.g. black) in order to reduce optical interference between neighboring subpixels. In this case, the bank **128** includes a light-blocking material made of at least one of a color pigment, organic black, or carbon.

[0037] The at least one light-emitting stack **124** is formed on the anode electrode **122** in a light-emitting area defined by the bank **128**. The at least one light-emitting stack **124** is formed by stacking a hole-related layer, an organic light-emitting layer, and an electron-related layer on the anode electrode **122** in that order or in the reverse order. In addition, the light-emitting stack **124** may include first and second light-emitting stacks that are opposite each other in the state in which a charge generation layer is disposed therebetween. In this case, the organic light-emitting layer of one of the first and second light-emitting stacks generates blue light, and the organic light-emitting layer of the other of the first and second light-emitting stacks generates yellowish-green light. Consequently, white light is generated by the first and second light-emitting stacks. The white light generated by the light-emitting stack **124** is incident on a color filter, which is located above or under the light-emitting stack **124**, to realize a color image. Alternatively, each light-emitting stack **124** may generate colored light corresponding to a respective subpixel without a separate color filter to realize a color image. That is, the light-emitting stack **124** of the red (R) subpixel may generate red light, the light-emitting stack **124** of the green (G) subpixel may generate green light, and the light-emitting stack **124** of the blue (B) subpixel may generate blue light.

[0038] The cathode electrode **126** is formed to be opposite the anode electrode **122** in the state in which the light-emitting stack **124** is disposed therebetween, and is connected to a low-voltage (VSS) supply line.

[0039] The encapsulation unit **140** reduces external moisture or oxygen from permeating into the light-emitting element **120**, which has low resistance to external moisture or oxygen.

[0040] To this end, the encapsulation unit **140** includes at least one of inorganic encapsulation layer **142** and inorganic encapsulation layer **146** and at least one organic encapsulation layer **144**. In the present disclosure, an encapsulation unit **140** having a structure in which a first inorganic encapsulation layer **142**, an organic encapsulation layer **144**, and a second inorganic encapsulation layer **146** are sequentially stacked will be described by way of example.

[0041] The first inorganic encapsulation layer **142** is formed on the substrate **111**, on which the cathode electrode **126** is formed. The second inorganic encapsulation layer **146** is formed on the substrate **111**, on which the organic encapsulation layer **144** is formed, and is formed to surround the upper surface, the lower surface, and the side surface of the organic encapsulation layer **144** together with the first inorganic encapsulation layer **142**.

[0042] The first and second inorganic encapsulation layers **142** and **146** reduce or prevent external moisture or oxygen from permeating into the light-emitting stack **124**. Each of the first and second inorganic encapsulation layers **142** and **146** is made of an inorganic dielectric material that can be deposited at a low temperature, such as silicon nitride (SiN.sub.x), silicon oxide (SiO.sub.x), silicon oxide nitride (SiON), or aluminum oxide (Al.sub.2O.sub.3). Consequently, each of the first

and second inorganic encapsulation layers **142** and **146** is deposited in a low-temperature atmosphere, whereby it is possible to reduce damage to the light-emitting stack **124**, which has low resistance to a high-temperature atmosphere, when each of the first and second inorganic encapsulation layers **142** and **146** is deposited.

[0043] Each of the first and second inorganic encapsulation layers **142** and **146** is formed to have a larger thickness than the passivation film **108**, whereby each of the first and second inorganic encapsulation layers **142** and **146** has low resistance to external impact. For this reason, each of the first and second inorganic encapsulation layers **142** and **146** is formed so as not to be disposed in a bending area BA.

[0044] The organic encapsulation layer **144** reduces stress between the layers due to bending of the organic light-emitting device and improves planarization. The organic encapsulation layer **144** is formed on the substrate **111**, on which the first inorganic encapsulation layer **142** is formed, and is made of a non-photosensitive organic dielectric material, such as a particle cover layer (PCL), an acrylic resin, an epoxy resin, polyimide, polyethylene, or silicon oxycarbide (SiOC), or a photosensitive organic dielectric material, such as photo acrylic. The organic encapsulation layer **144** is disposed in an active area AA, excluding a non-active area NA. At this time, a dam **106** is disposed on the substrate **111** or the passivation film **108** in order to prevent the organic encapsulation layer **144** from spreading to the non-active area NA.

[0045] A touch-sensing line **154** and a touch-driving line **152** are disposed in the active area AA of the encapsulation unit **140** to intersect each other in the state in which a touch dielectric film **156** is disposed therebetween. Mutual capacitance C_m is formed at the intersection of the touch-sensing line **154** and the touch-driving line **152**. Consequently, the mutual capacitance C_m charges an electric charge by a touch-driving pulse supplied to the touch-driving line **152** and discharges the charged electric charge to the touch-sensing line **154**, thereby serving as a touch sensor.

[0046] The touch-driving line **152** includes a plurality of first touch electrodes **152e** and first bridges **152b** for electrically interconnecting together the first touch electrodes **152e**.

[0047] The first touch electrodes **152e** are spaced apart from each other on the touch dielectric film **156** by a predetermined distance in a Y direction, which is a first direction. Each of the first touch electrodes **152e** is electrically connected to an adjacent first touch electrode **152e** via a corresponding one of the first bridges **152b**.

[0048] The first bridges **152b** are disposed on the touch dielectric film **156**, which is disposed in the same plane as the first touch electrodes **152e**, so as to be electrically connected to the first touch electrodes **152e** without separate contact holes. The first bridges **152b** are disposed to overlap the bank **128**, whereby it is possible to reduce the reduction of an aperture ratio due to the first bridges **152b**.

[0049] The touch-sensing line **154** includes a plurality of second touch electrodes **154e** and second bridges **154b** for electrically interconnecting the second touch electrodes **154e**.

[0050] The second touch electrodes **154e** are spaced apart from each other on the touch dielectric film **156** by a predetermined distance in an X direction, which is a second direction. Each of the second touch electrodes **154e** is electrically connected to an adjacent second touch electrode **154e** via a corresponding one of the second bridges **154b**.

[0051] The second bridges **154b** are formed on a touch buffer layer **148**, and are electrically connected to the second touch electrodes **154e** via touch contact holes **150**, which are formed through the touch dielectric film **156**. In the same manner as the first bridges **152b**, the second bridges **154b** are disposed to overlap the bank **128**, whereby it is possible to reduce the reduction of an aperture ratio due to the second bridges **154b**.

[0052] Meanwhile, the structure in which the second bridges **154b** are disposed on the touch buffer layer **148** to contact the touch buffer layer **148** has been described by way of example with reference to FIG. 3. Alternatively, at least one of the first and second touch electrodes **152e** and **154e** or the first bridges **152b** may be disposed on the touch buffer layer **148** to contact the touch

buffer layer **148**, and the second bridges **154b** may be disposed on the touch dielectric film **156**. [0053] In addition, each of the first bridges **152b**, the second bridges **154b**, the first touch electrodes **152e**, and the second touch electrodes **154e** has been described by way of example as being formed in a plate shape, as shown in FIG. 2. Alternatively, at least one of the first bridges **152b**, the second bridges **154b**, the first touch electrodes **152e**, or the second touch electrodes **154e** may be formed in a mesh shape.

[0054] In the present disclosure, the structure in which the first touch electrodes **152e** and the second touch electrodes **154e** are formed in a mesh shape will be described by way of example with reference to FIGS. 4 and 5. The mesh-shaped touch electrodes **152e** and **154e** correspond to the bank **128** disposed in a non-emission area NEA of each subpixel, and an opening area between the mesh-shaped touch electrodes **152e** and **154e** corresponds to an emission area EA of each subpixel. Each of the touch electrodes **152e** and **154e** includes a transparent conductive film, such as ITO or IZO, and a mesh metal film disposed above or under the transparent conductive film, the mesh metal film being formed in a mesh shape. Alternatively, each of the touch electrodes **152e** and **154e** may include only a mesh metal film. Here, the mesh metal film has a structure having at least one layer made of at least one of Ti, Al, Mo, MoTi, Cu, Ta, or ITO, which exhibits higher conductivity than the transparent conductive film, and is formed in a mesh shape. For example, the mesh metal film is formed to have a three-layer stack structure, such as Ti/Al/Ti, MoTi/Cu/MoTi, or Ti/Al/Mo. Consequently, the resistance and capacitance of each of the first and second touch electrodes **152e** and **154e** and the first and second bridges **152b** and **154b** are reduced, whereby an RC time constant is reduced and thus touch sensitivity is improved. In addition, the line width of the mesh metal film of each of the first and second touch electrodes **152e** and **154e** is very small, whereby it is possible to prevent the reduction of an aperture ratio and transmittance due to the mesh metal film.

[0055] Meanwhile, a display pad **168**, which is connected to at least one of the data line DL, the scan line SL, the low-voltage (VSS) supply line, or the high-voltage (VDD) supply line, and a touch pad **160** are disposed in the non-active area NA. The display pad **168** and the touch pad **160** may be disposed in the portion of the non-active area NA corresponding to at least one of a one-side portion or an other-side portion of the substrate **111**, or may be disposed in different portions of the non-active area NA. Meanwhile, the disposition of the touch pad **160** and the display pad **168** is not limited to the structure shown in FIG. 2. The disposition of the touch pad **160** and the display pad **168** may be variously changed depending on the design of the display panel.

[0056] The touch pad **160** and the display pad **168** are disposed on a dielectric film disposed under the light-emitting element **120**. For example, the touch pad **160** is disposed on the substrate **111** together with the display pad **168**. The touch pad **160** and the display pad **168** are formed to be exposed by a touch passivation film **158**. Consequently, the touch pad **160** is connected to a signal transmission film having a touch-driving circuit (not shown) mounted thereon, and the display pad **168** is connected to a signal transmission film having at least one of a scan-driving unit or a data-driving unit mounted thereon. Meanwhile, the touch-driving circuit may be mounted in one of the data-driving unit and a timing controller.

[0057] The touch pad **160** includes first pad electrode **162** and second pad electrode **164**, which are electrically connected to each other.

[0058] The first pad electrode **162** is made of the same material as the source and drain electrodes **136** and **138**, and is disposed on the substrate **111**. Meanwhile, the first pad electrode **162** may be made of the same material as the source and drain electrodes **136** and **138**, and may be disposed in the same plane as the source and drain electrodes **136** and **138**.

[0059] The second pad electrode **164** is made of the same material as a routing line **170**, and is disposed on the touch dielectric film **156**. The second pad electrode **164** is connected to the first pad electrode **162**, which is exposed through a pad contact hole **166** formed through the touch dielectric film **156** and the touch buffer layer **148**. Meanwhile, the routing line **170**, which is

connected to the second pad electrode **164**, extends from each of the touch electrodes **152e** and **154e**, and is formed along the side surface of the encapsulation unit **140**. At this time, the routing line **170** is disposed so as to cross at least one dam **106**. The routing line **170** is made of the same material as the touch electrodes **152e** and **154e**, and is formed on the touch dielectric film **156**. The routing line **170** overlaps a black layer **184** and a white layer **182**, which are disposed on the touch passivation film **158**, so as to reduce external light from being reflected by the routing line **170**.

[0060] The touch passivation film **158** are formed to cover the touch sensor, which includes the touch electrodes **152e** and **154e** and the bridges **152b** and **154b**, in order to reduce the touch sensor from being corroded by external moisture. In addition, the touch passivation film **158** is made of an organic dielectric material, such as epoxy or acrylic, and is formed in the form of a thin film or a general film, or is made of an inorganic dielectric material, such as SiN.sub.x or SiO.sub.x.

[0061] As shown in FIGS. **5** and **6**, an antireflective film **180** including a white layer **182** and a black layer **184** is disposed on the touch passivation film **158**.

[0062] The black layer **184** reduces mixing of internal light generated by the light-emitting element **120** and absorbs external light incident from the outside. The black layer **184** is formed in a mesh (matrix) shape so as to overlap the mesh-shaped first and second touch electrodes **152e** and **154e** and the bank **128**. That is, the black layer **184** is formed in the emission area EA, but is not formed in the non-emission area NEA. The black layer **184** may be formed on the white layer **182** so as to have the same line width as the white layer **182** in order to expose the side surface of the white layer **182**, as shown in FIG. **5**, or may be formed on the white layer **182** so as to have a larger line width than the white layer **182** in order to cover opposite side surfaces of the white layer **182**, as shown in FIG. **6**.

[0063] The black layer **184** includes black nanoparticles, a binder, and a sensitive emulsion. The black nanoparticles include at least one of carbon-black-based black nanoparticles, metal-oxide-based black nanoparticles, or organic-based black nanoparticles. TiN_xO_y or CuMnFeO_x is used as the metal-oxide-based black nanoparticles, and lactam black or perylene black is used as the organic-based black nanoparticles.

[0064] The white layer **182** is formed so as to overlap the black layer **184** while contacting the black layer **184** between the black layer **184** and the touch sensor. The white layer **182** is formed on the touch passivation film **158** in a mesh (matrix) shape so as to overlap the mesh-shaped first and second touch electrodes **152e** and **154e** and the bank **128**. That is, the white layer **182** is formed in the emission area EA, but is not formed in the non-emission area NEA. The white layer **182** includes white nanoparticles, a binder, and a sensitive emulsion. The white nanoparticles include at least one of Ti-based white nanoparticles, silica-based white nanoparticles, metal-oxide-based white nanoparticles, or core-shell-based white nanoparticles. TiO₂ is used as the Ti-based white nanoparticles, spherical SiO₂, hollow SiO₂, fumed SiO₂, or colloidal SiO₂ is used as the silica-based white nanoparticles, ZrO₂, Al₂O₃, or CeO₂ is used as the metal-oxide-based white nanoparticles, and a silica-based core and a Ti-based shell are used as the core-shell-based white nanoparticles. As shown in FIG. **7**, the higher the content of the white nanoparticles included in the white layer **182**, the higher the reflectance. Consequently, the white nanoparticles may be included in the white layer **182** at a content of 90% to 97%.

[0065] As shown in FIG. **8**, the white layer **182** changes the path of the internal light IL, generated by the light-emitting stack **124** and directed toward the non-emission area NEA, to the emission area EA. That is, the white layer **182** may reflect the internal light IL advancing to the non-emission area NEA to the emission area EA, whereby it is possible to improve the efficiency of the light that exits through the emission area EA.

[0066] A process of reducing the reflection of external light using the black layer **184** and the white layer **182** according to the present disclosure will be described with reference to FIG. **8**. Most external light OL is absorbed and extinguished by the black layer **184**. However, some of the external light OL is not absorbed by the black layer **184** but is reflected from the black layer **184** as

a first reflection light beam L1. In addition, some of the external light absorbed by the black layer **184** is transmitted through the black layer **184** and is reflected from the black layer **184** as a second reflection light beam L2. At this time, the first and second reflection light beams L1 and L2 have opposite phases, whereby the first and second reflection light beams L1 and L2 destructively interfere with each other and thus are extinguished.

[0067] In the present disclosure, as described above, the external light is absorbed using the black layer **184**, and the external light is extinguished through destructive interference using the black layer **184** and the white layer **182**, whereby it is possible to prevent the reflection of the external light. In particular, as shown in Table 1, the reflectance of external light is 6.5% according to Comparative Example 1, in which a low-reflective metal is used, and the reflectance of external light is 5.3% according to Comparative Example 2, in which a black layer is used, whereas the reflectance of external light is less than 5.0%, i.e. 1.6%, according to Example, in which the black layer **184** and the white layer **182** are used. Consequently, the reflectance of external light according to Example is lower than the reflectance of external light according to each of Comparative Examples 1 and 2.

TABLE-US-00001

TABLE 1	Comparative Example 1	Example 2	Example
Reflectance	6.5%	5.3%	1.6%

[0068] In the present disclosure, as described above, it is possible to reduce external light from being incident on the touch sensor and the routing line **170** using the black layer **184** and the white layer **182**, whereby it is possible to reduce the reflectance of the external light. In the present disclosure, therefore, it is possible to reduce a reduction in external visibility, and therefore it is possible to obviate a polarizing plate, which is expensive. Also, in the present disclosure, because the expensive polarizing plate is obviated, it is possible to improve transmittance and luminance. Consequently, the power consumption thereof is reduced, and the lifespan thereof is increased. Meanwhile, the substrate **111**, on which the black layer **184** and the white layer **182** according to the present disclosure are formed, is laminated with a cover substrate **101** via an adhesive film **186**, as shown in FIGS. **9A** and **9B**. The adhesive film **186** is formed between the black layer **184** and the cover substrate **101** in the form of an optical clear adhesive (OCA) or an optical clear resin (OCR). The adhesive film **186** and the black layer **184** may contact each other, as shown in FIG. **9A**, or an optical film **188** may be disposed between the adhesive film **186** and the black layer **184**, as shown in FIG. **9B**. A semi-transmissive film or an OLED transmittance controllable film (OTF) is used as the optical film **188**.

[0069] A display device comprising a display panel with touch sensors shown in FIG. **1** and FIG. **2** is further disclosed according to another embodiment of the present invention. And the same description for the display panel with touch sensors in the display device will be omitted here.

[0070] As is apparent from the above description, in the present invention, it is possible to reduce external light from being incident on the touch sensor and the cathode electrode using the antireflective film (for example, the black layer and the white layer), whereby it is possible to reduce the reflectance of external light. In the present disclosure, therefore, it is possible to reduce a reduction in external visibility without using a separate optical film (e.g. a polarizing plate). Also, in the present disclosure, it is possible to improve the transmittance, luminance, and lifespan of the display panel and the display device. Consequently, it is possible to reduce the thickness and the power consumption of the display panel and the display device.

[0071] It will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the spirit or scope of the invention. Thus, it is intended that the present invention covers such modifications and variations to this invention provided they come within the scope of the appended claims and their equivalents.

Claims

- 1.** A display device comprising: a light-emitting element on a substrate, the light-emitting element comprising an anode electrode, at least one light-emitting stack on the anode electrode, and a cathode electrode on the at least one light-emitting stack; a bank on the substrate, the bank exposing the anode electrode of the light-emitting element; an encapsulation unit on the light-emitting element and the bank, the encapsulation unit including at least one inorganic encapsulation layer and at least one organic encapsulation layer; a touch sensor on the encapsulation unit; and a black layer on the touch sensor, the black layer overlapping the bank.
 - 2.** The display device according to claim 1, wherein the black layer is on a second area adjacent to a first area in which the anode electrode exposed from the bank is disposed.
 - 3.** The display device according to claim 2, wherein the bank includes a light-blocking material.
 - 4.** The display device according to claim 2, wherein the black layer overlaps the touch sensor.
 - 5.** The display device according to claim 4, wherein the touch sensor comprises: first touch electrodes arranged in a first direction, the first touch electrodes connected to each other via a first bridge; and second touch electrodes arranged in a second direction that is different from the first direction, the second touch electrodes connected to each other via a second bridge, wherein the first touch electrodes and the second touch electrodes include metal lines that form a mesh shape.
 - 6.** The display device according to claim 5, wherein the black layer overlaps the metal lines.
 - 7.** The display device according to claim 6, wherein the black layer has opening areas and the opening areas overlap the first area.
 - 8.** The display device according to claim 5, wherein the first touch electrodes and the second touch electrodes have opening areas, respectively, and wherein the opening areas of the first touch electrodes and the opening areas of the second touch electrodes overlap the first area.
 - 9.** The display device according to claim 5, wherein the first bridge and the second bridge overlap the second area.
 - 10.** The display device according to claim 9, wherein the first bridge is on a same layer as the first touch electrodes and the second touch electrodes, and the second bridge is on a different layer from the first bridge.
 - 11.** The display device according to claim 1, further comprising: a color filter located above the at least one light-emitting stack.
 - 12.** The display device according to claim 1, further comprising: a touch passivation film between the touch sensor and the black layer.
 - 13.** The display device according to claim 12, wherein the touch passivation film comprises an inorganic material.
 - 14.** The display device according to claim 13, wherein the inorganic material includes a silicon oxide layer or a silicon nitride layer.
 - 15.** The display device according to claim 1, wherein the black layer includes at least one of black nanoparticles, a binder, or a sensitive emulsion.
 - 16.** The display device according to claim 15, wherein the black nanoparticles include at least one of carbon-black-based black nanoparticles, metal-oxide-based black nanoparticles, or organic-based black nanoparticles.
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